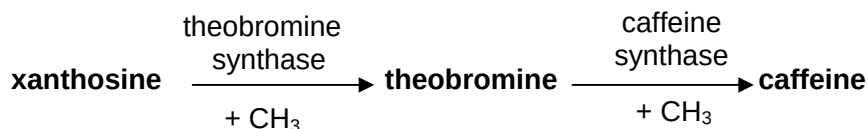




MERIDIAN JUNIOR COLLEGE
Preliminary Examinations 2011
H2 BIOLOGY 9648/03 Paper 3 - Answers

1. The synthesis of caffeine in coffee plants involves enzymes which add methyl groups (CH_3) to convert xanthosine to caffeine:



In an attempt to produce caffeine-free coffee, cells of a coffee plant, *Coffea canephora*, were grown in tissue culture and genetically modified to suppress expression of the gene for theobromine synthase.

DNA was constructed to code either for short or for long lengths of RNA with base sequences complementary to parts of the mRNA produced by the gene for theobromine synthase.

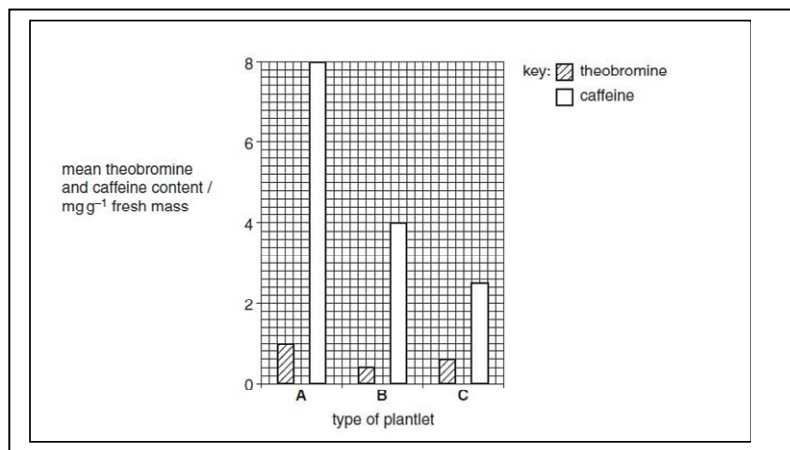
- (a) Explain how lengths of RNA that are complementary to mRNA may suppress the expression of the gene for theobromine synthase. [3]

- **mRNA and its complementary RNA bind together**
- **by hydrogen bonding**
- **A to U and C to G**
- **form double stranded RNA / duplex RNA**
- **cannot bind to ribosome**
- **cannot be translated**
- **tRNA cannot bind**

Three types of cell were then cloned in tissue culture into plantlets:

- A** - unmodified cells
- B** - genetically modified cells with DNA that codes for short lengths of RNA complementary to mRNA for theobromine synthase
- C** - genetically modified cells with DNA that codes for long lengths of RNA complementary to mRNA for theobromine synthase.

Samples of each of the three types of plantlet were analysed to measure their theobromine and caffeine content. The results of the analysis are shown in Fig. 1.1.



(b) With reference to Fig. 1.1, describe and explain the results of the experiment. [5]

Description of data: max3 (for both theobromine and caffeine mentioned)

When plants were genetically modified with DNA that codes for short / long length of RNA,

- **Theobromine content reduced from ___ to ___ + appropriate figures [1]**
- **Slight difference between short and long lengths of RNA + appropriate figures [1]**
- **Caffeine content reduced from ___ to ___ + appropriate figures [1]**

Explanation:

- **Long length RNA has a higher chance of base pairing with mRNA**
- **Hence, reduced translation into theobromine synthase**
- **With less enzyme present, less theobromine formed for conversion to caffeine**

(c) Describe briefly how plants are cloned by tissue culture. [4]

- **explant of meristematic/ totipotent / pluripotent cells or tissues obtained**
- **surface sterilised with sodium hypochlorite solution**
- **placed on nutrient medium containing appropriate growth regulator to stimulate mitosis / division**
- **callus formed**
- **subcultured in medium containing appropriate growth regulator to stimulate differentiation /**
- **formation of roots and shoots / plantlet**
- **transfer to soil to allow further growth**

(d) Explain the advantages of using cloned plants in experiments such as this. [3]

- **Genetically identical [1]. Hence, genotype does not affect result [1]**
- **Large numbers easily obtained**
- **Plantlets / callus are compact [1], hence easily kept in identical conditions [1]**
- **Can be obtained independent of growing seasons**

[Total: 15]

2. Huntington's disease (HD) is a lethal neurodegenerative disease that is inherited as an autosomal dominant condition. As the appearance of symptoms may occur later in life (as late as at age 40), patients with HD may unknowingly pass on the mutant HD allele to their children.

A radioactive G8 probe revealed two RFLP variants that are linked to the HD allele which can be used in the screening of HD. Fig. 2.1 shows the two RFLP variants upstream to the HD allele and the G8 probe. The arrows show the position of *Hind*III restriction sites. The numbers represent kilobases.

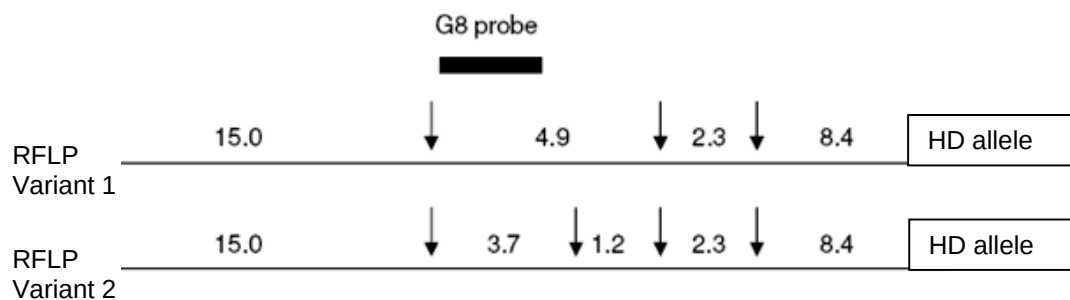


Fig. 2.1

Fig. 2.2 illustrates the Southern Blot analysis of the two RFLP variants in 30 related individuals with a family history of HD.

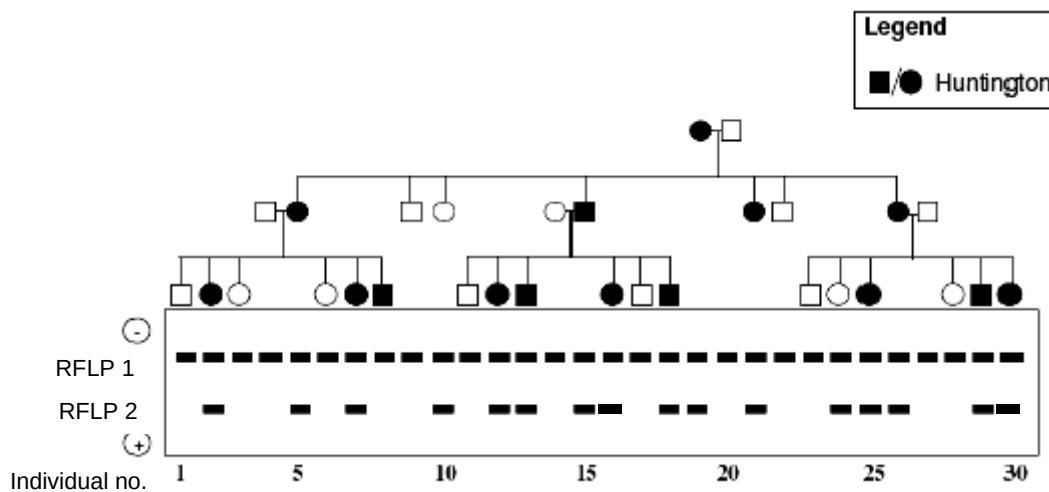


Fig. 2.2

With reference to Fig. 2.1 and 2.2,

(a) outline how the DNA band patterns of the 30 individuals were obtained. [6]

1. DNA sample from the 30 individuals is treated/digested with *Hind*III (into restriction fragments)
2. Restriction fragments are subjected to gel electrophoresis where the fragments are separated according to sizes.
3. Shorter fragments move further OR cite figures (eg. 1.2kb fragments move further than the 4.9kb fragments)

4. The separated DNA bands are transferred to nitrocellulose membrane by Southern Blotting
5. Double stranded DNA denatured by NaOH / alkali
6. Flushed with solution of G8 probe which is complementary to / which binds to / anneals to / hybridizes with / forms hydrogen bonds with 4.9kb and 3.7kb fragments
7. Viewed by autoradiography / radioactivity exposes photographic film

(b) identify the RFLP variant which is linked to the mutant HD allele. Explain your answer. [2]

- **RFLP 2**

Explanation:

- Most affected individuals have RFLP variant 2 except individual 8
- Most unaffected individuals do not have RFLP variant 2 except individuals 10, 24
- Most affected individuals have a shorter 3.7kb fragment except individual 8
- Most unaffected individuals have a 4.9kb fragment except individual 24

(c) suggest two possible reasons for the difference in the DNA band patterns among unaffected individuals. [4]

- Individual 10 and 24 received from affected parent a chromosome with mutant HD allele linked to RFLP 1...
- ...although the normal HD allele is linked to RFLP 1

OR

- Mutation in a HindIII restriction site of RFLP 1 / in 4.9kb fragment led to...
- ... RFLP1 becoming RFLP2 (OWTTE) / appearance of one more HindIII restriction site. Hence, variant 1 not detected in individuals 10 & 24.

OR

- During the formation of ovum in the female parent of individual 10 and 24, crossing over occurred at a region between the RFLP locus and HD locus...

- ... cause RFLP2 to become associated with normal allele

OR

- Individual 10 and 24 have the mutant allele associated with RFLP 2 but symptoms have not appeared
- ... as disease occur later in life / as late as 40 years / person still young

(d) discuss the effectiveness of using this RFLP marker in the screening of HD. [2]

- not very effective / may lead to wrong diagnosis
- the linkage between RFLP and the HD allele is not close enough to prevent crossing over from occurring

[Total: 14]

3.

(a) Outline the important features and the normal role of embryonic stem cells at the 4-cell stage in living organisms. [3]

- Has full set of genetic information
- Totipotent
- able to differentiate into or give rise to all cells in the organism
- Forms the fetus (OWTTE)

(b) Heart attack causes damage to heart muscles. To repair damaged heart tissues, bone marrow stem cells are harvested from the patient, cultured and injected directly into the heart during coronary by-pass operation.

(i) Describe an important feature of the bone marrow stem cells that enables this to happen. [2]

- show plasticity in developmental potential / able to transdifferentiate
- have potential to differentiate into other cell types

(ii) Suggest how the bone marrow stem cells are coaxed to become healthy heart muscle tissue. [2]

- Chemical signal molecules secreted by the heart cells / Physical contact with surrounding heart cells
- Switched on the genes for the development of heart cells

(iii) Suggest why the bone marrow stem cells from the patient but not from other donors were used to repair the damaged heart muscles. [1]

- No tissue rejection

(c) About twenty percent of the DNA produced by PCR is replicated inaccurately. Suggest and explain why it is not effective to use CFTR gene cloned by PCR in treating cystic fibrosis. [3]

- percentage risk is too high for human application
- Taq DNA polymerase do not have proofreading mechanism
- High proportion of mutated / not all DNA produced will be the normal CFTR allele
- Points that suggest problem not solved

[Total: 11]

Planning Question

4. DNA is the genetic material in all living organisms. DNA from an external source can be taken up by bacteria by a process known as transformation.

Plan an experiment to investigate the effect of varying plasmid concentration on the efficiency of *E. coli* transformation.

You may, or otherwise, also use the following equipment and materials:

- Solution of plasmid containing ampicillin resistance gene of concentration 100 Unit/ml
- Suspension of competent *E. coli* cells
- Distilled water
- LB broth (nutrient solution)
- Petri dishes containing LB agar medium and ampicillin (LB/amp plates)
- Sterilised microfuge tubes
- Micropipettes
- Bleach solution
- alcohol

Your plan should have a clear and helpful structure to include:

- an explanation of theory to support your practical procedure
- a description of the method used including scientific reasoning behind the method
- an explanation of the dependent and independent variables involved
- relevant, clearly labelled diagrams
- how you will record your results and ensure they are as accurate and reliable as possible
- proposed layout of results tables and graphs with clear headings and labels
- correct use of scientific and technical terms

[Total:12]

Mark Scheme

1. Theoretical consideration / rationale of plan to justify the practical procedure [1]
2. Correct use of technical and scientific terms – any 2 of the following [1]

Relate theory to prac

Examples:

- DNA is the genetic material in all living organisms. Transformation is the process by which DNA from an external source is taken up by bacteria.
- The plasmid contains an ampicillin-resistance gene that confers resistance to the antibiotic.
- Bacteria that have taken up this plasmid are able to grow on LB/amp medium.
- Amp resistance gene serves as genetic marker to identify transformed cells
- When made competent by heat-shock, pores appear transiently in the membrane of bacteria. External DNA/plasmids may then enter the cell via these pores

- Consequently, as plasmid concentration increases, the efficiency of *E.coli* transformation will also increase proportionally.

3. An explanation of the dependent and independent variables involved [1]

- Independent variable: Concentration of plasmid (unit/ml)
- Dependent variable: Efficiency of *E. coli* transformation is represented by Number of colonies on LB/Amp plates

[A description of method used including the scientific reasoning behind the method]

4. Description of how the plasmid solution is prepared [1]

- Using plasmid solution of concentration 100 Unit/ml, make dilutions to obtain 5 different concentrations.

Concentration of plasmid (unit/ml)*	Volume of plasmid solution of conc 100 Units/ml (ml)*	Volume of water (ml)*
100	1.0	0.0
80	0.8	0.2
60	0.6	0.4
40	0.4	0.6
20	0.2	0.8
0	0	1.0

* or any other reasonable range / volume

OR

serial dilutions:

Concentration of plasmid (unit/ml)	Volume of stock (ml)	Volume of water (ml)
Undiluted	1.0	0.0
1 to 10 or 10^{-1}	0.1 of non dilute	0.9
1 to 100 or 10^{-2}	0.1 of 10^{-1}	0.9
1 to 1000 or 10^{-3}	0.1 of 10^{-2}	0.9
1 to 10000 or 10^{-4}	0.1 of 10^{-3}	0.9

5. Controlling factors that affect experiment (eg. Duration, temp, volumes,etc) [1]

6. Description of expt [1]

- Mix a fixed volume of the *E. coli* culture with a fixed volume (eg 10µl) of the each plasmid solution
- Heat-shock the bacteria-plasmid mixture
- Plate each sample on LB/amp plate.
- Seal the plates with tape and incubate for 24 hours in a 37°C (range 28-37) incubator oven.
- After an overnight incubation, count the number of colonies in each plate.

[How you will record your results and ensure they are as accurate and reliable as possible]

7. Control [1]

- vii. For the **control** tube, replace plasmid solution with same volume of 10 μ l of distilled water/buffer solution.

Rationale:

To show that the presence of bacterial colonies on agar plates containing ampicillin is due to transformed bacteria which have taken up plasmid with amp resistance gene OR
in the absence of plasmid which carries the ampicillin resistant genes, there are no ampicillin resistant bacterial cells.

8. Accuracy and reliability [1]

- For each plasmid concentration, perform two replicates / repeat steps ___ to ___ twice
- Calculate average
- Check for at least 5 different concentrations of plasmid solutions

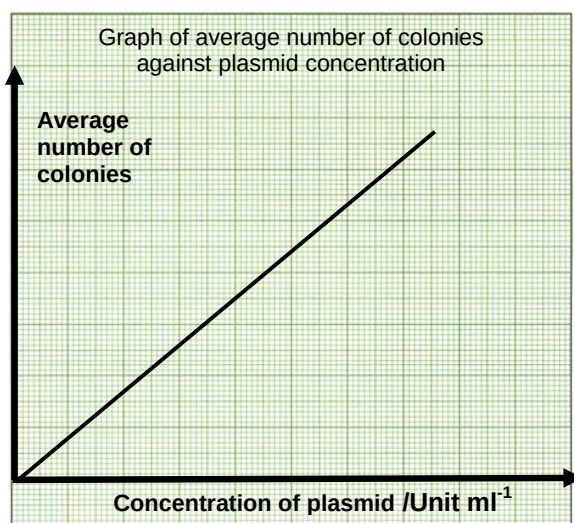
[Propose layout of results table and graph with clear headings and labels]

9. Table with appropriate headings [1]

Concentration of plasmid solution (unit/ml)	Number of colonies in each plate			
	Reading 1	Reading 2	Reading 3	Average

10. Graph [1]

- Clear label of axes with units and consistent headings of table
- propose expected trend



11. Risk Assessment [1] – any 2

- E coli* is a bacteria and can cause disease.

Safety Precautions:

- Avoid contact - wear gloves and safety goggles.
- If in contact, wash off with water and antibacterial soap immediately.
- If skin is broken, obtain medical attention.
- Dispose of used microfuge tubes / pipette tips / inoculation loops in bleach solution
- Sterilize bench with alcohol after expt

12. Suitable diagram illustrating plates with bacteria / process of heat shock [1]

5. (a) The most common form of severe combined immunodeficiency disease (SCID) is caused by an X-linked mutation.

(i) Explain how X-linked SCID may be treated using a viral delivery system. [8]

1. The RNA copies of normal allele is obtained from bone marrow of donor.
2. Both viral RNA and RNA of normal gene are cut with the same restriction enzyme to produce the same sticky ends.
3. ... annealed and join by RNA ligase to produce recombinant RNA molecule.
4. The recombinant RNA molecule is packaged into modified retrovirus vector which does not cause disease.
5. Harvest haemopoietic stem cells / bone marrow stem cells / T cells from patient
6. The modified retrovirus is used to infect harvested cells (ex vivo on a petri dish).
7. Reverse transcription / transcriptase → double-stranded copies of normal gene are created
8. These DNA copies are integrated randomly into the host chromosome/genome/DNA
9. The stem cells / T cells are transplanted back into the body of the patient
10. where they divide and differentiate to produce normal lymphocytes (T cells, B cells).

(ii) Discuss the advantages and limitations of treatment using viral delivery method. [5]

1. Disadv - Treatment through modifying T cells is not permanent/had to be repeated
2. ... because of limited life span of T cells (6–12 months)
3. Adv: Treatment through modifying stem cells is more permanent
4. ... as stem cells divide continuously
5. Disadv - The integration of normal allele into host genome by retrovirus is random.
6. This may disrupt normal functions of host genes,

7. ...resulting in the development of cancer or leukaemia
8. Random, therefore integrated into region where normal allele not well expressed

(b) Discuss the benefits of the Human Genome Project (HGP) to human health. [7]

1. HGP identified all the approximately 20,000-25,000 genes in human DNA
2. The entire human genome sequenced (3 billion base pairs)
3. This detailed genome map helped researchers identify genes associated with genetic conditions
4. Examples include inherited colon cancer genes, Alzheimer's disease and familial breast cancer (BRCA1 , BRCA2 genes)

Diagnosis

5. Allowed for development of more specific diagnostic tests
6. Hence, earlier detection / diagnosis of diseases or genetic predisposition to diseases
7. With genetic screening/testing, the likelihood of inheriting genetic diseases may be reduced through genetic counseling.
8. By comparing the genome between humans and other organisms (eg. mice) has led to discovery of similar genes associated with diseases and traits.

Treatment of diseases

9. Drugs may now be designed to target fundamental genetic causes of diseases rather than treatment of symptoms
10. Drugs may be tailor-made for individuals and adapted to each person's own genetic makeup → greater efficacy and safety
11. Create drugs based on the proteins, enzymes, and RNA molecules associated with genes and diseases.
12. Facilitate drug discovery and allow drug makers to produce a therapy more targeted to specific diseases
13. → maximize therapeutic effects and decrease damage to healthy cells.
14. Makes replacement of defective genes through gene therapy easier.

Risk assessment

15. With the knowledge that genetic differences make some **people more susceptible** and others **more resistant to agents** such as radiation exposure, mutagenic chemicals and cancer causing chemicals, **risks posed to individuals caused by exposure to such agents** may be assessed
